

CS 255 Model Application Short Paper

ERNEST KOFI LOGAH

ernest.logah@snhu.edu

Southern New Hampshire University

Process Model Application

A process model focuses on the flow of activities performed within DriverPass.

For this project, process modeling would be used to document customer registration, package selection, lesson scheduling, online testing, progress tracking, and reporting.

By mapping these workflows, DriverPass can understand how information moves through the system and identify potential bottlenecks or inefficiencies.

Process models are particularly useful because DriverPass serves multiple stakeholders **(Dennis et al., 2020)**. Customers, secretaries, instructors, management, and IT administrators all interact with the system differently. A process model helps ensure that each activity is performed in the correct sequence and that business rules are consistently applied.

Object Model Application

An object model focuses on the entities that make up the DriverPass system and the relationships among them.

Examples of objects include Customer, Instructor, Vehicle, Lesson, Reservation, Package, Practice Test, User Account, and Report.

Object modeling would help developers design the software structure by defining attributes

and behaviors for each object **(Dennis et al., 2020)**. For example, a Customer object may contain personal information and lesson history, while a Reservation object stores scheduling information. This approach improves maintainability, reusability, and scalability.

Process and Object Model Comparison

Both modeling approaches provide value to the DriverPass project **(Dennis et al., 2020)**. Process models excel at showing workflows and illustrating how users interact with the system. They are useful for understanding business operations and identifying inefficiencies. However, process models provide limited information about software structure.

Object models excel at representing the data and entities required by the system. They support software design, maintenance, and future enhancements. However, they do not always show the complete flow of business activities as clearly as process models.

For DriverPass, the strongest solution is to use both approaches together. Process models help stakeholders understand how the business operates, while object models provide the technical foundation needed to build a reliable and maintainable system.

References

Dennis, A., Wixom, B. H., & Tegarden, D. (2020). Systems analysis and design: An object-oriented approach with UML (6th ed.). Wiley.